

Problem 1

((c) $\implies$ (a))

Assume (c)

Let $\varphi : G \rightarrow H \times K$ s.t. $\varphi(g) = (\phi'(g), \psi(g))$

Homomorphism:

Since ϕ' and ψ are homomorphisms φ is a homomorphism.

Injective:

Let $g \in \ker(\varphi)$

Thus $\varphi(g) = (\phi'(g), \psi(g)) = (e, e)$ thus $g \in \ker(\phi'), \ker(\psi)$ thus since this is an exact sequence $\ker(\psi) = \text{im}(\phi)$ thus since $\phi' \circ \phi = \text{id}_H \implies g = e$ thus $\ker(\varphi) = \{e\}$

Surjective: Let $(h, k) \in H \times K$, thus since σ is surjective there exists a $g \in G$ s.t. $\psi(g) = k$ and for the h consider $\phi(h)$

Thus $\varphi(\phi(h), g) = (\phi'(\phi(h)), \psi(g)) = (h, k)$ thus surjective. Thus $G \cong H \times K$ thus proven in class since this is a direct product this sequence is split.

((a) $\implies$ (b))

Assume (a)

Thus shown in class this means there exists a $K' \cong K$ s.t. $G = \phi(H)K$ and $\phi(H) \cap K' = 1$

Consider $\psi|_{K'} : K' \rightarrow K$ is a homomorphism.

Injective:

Let $k' \in \ker(\psi|_{K'})$ thus $\psi|_{K'}(k') = e \in \phi(H) \implies k' = e$ since $\phi(H) \cap K' = \{e\}$

Surjective:

For any $k \in K$, since ψ is surjective, there exists $g \in G$ s.t. $\psi(g) = k$

Thus you can write $g = \phi(h)k'$ for $h \in H$ and $k' \in K'$

$k = \psi(g) = \psi(\phi(h)k') = \psi(\phi(h))\psi(k') = \psi(k')$

Thus $\psi|_{K'}$ is surjective.

Since $\psi|_{K'}$ is an isomorphism take its inverse and define $\psi' : K \rightarrow G$ s.t. $\psi' = (\psi|_{K'})^{-1}$ we can do this since $K' \subset G$

ψ' is a homomorphism since $(\psi|_{K'})^{-1}$ is a homomorphism.

Consider $\psi(\psi'(k))$ for $k \in K$ well as shown above ψ' is actually only going to map to K' and because of this we can think of $\psi = \psi|_{K'}$ thus since these are inverse of each other

$\psi(\psi'(k)) = k$ thus done

((b) $\implies$ (a))

Assume (b)

Define $K' = \psi'(K)$

Let $x \in \phi(H) \cap K'$

Thus $x = \phi(h) = \psi'(k)$ for $h \in H$ and $k \in K$

$\psi(\phi(h)) = \psi(\psi'(k)) \implies e = k \implies x = \psi'(e) = e$ thus $\phi(H) \cap K' = \{e\}$

Let $g \in G$

Define $k' = \psi'(\psi(g)) \in K'$

Consider $\psi(gk'^{-1}) = \psi(g)\psi(k'^{-1}) = \psi(g)\psi(k')^{-1} = \psi(g)\psi(\psi'(\psi(g)))^{-1} = \psi(g)\psi(g)^{-1} = e$

Thus $gk'^{-1} \in \ker(\psi) = \text{im}(\phi)$ thus there is an $h \in H$ s.t. $\phi(h) = gk'^{-1}$ thus $g = \phi(h)k'$

Thus $G = \phi(H)K'$

Thus G is split

Problem 2

For a group of order 8 to be split in this sequence $G \cong C_2 \rtimes C_2 \times C_2$ thus there exists a $\phi : C_2 \times C_2 \rightarrow \text{Aut}(C_2) \cong \{e\}$ thus the only homomorphism $\phi(a) = \text{id}_{C_2 \times C_2}$ s.t. $a \in C_2$ thus to be split $G \cong C_2 \times C_2 \times C_2$

1. C_8

If $N \cong C_2$, then $C_8/N \cong C_4$ thus C_8 cannot fit in this sequence.

2. $C_4 \times C_2$

$$C_2 \cong \langle (a^2, e) \rangle$$

$$C_4 \times C_2 / \langle (a^2, e) \rangle = \{ \langle (a^2, e) \rangle, (a, e) \langle (a^2, e) \rangle, (e, b) \langle (a^2, e) \rangle, (a, e) \langle (a^2, e) \rangle \} \cong C_2 \times C_2$$

$$e \rightarrow \langle (a^2, e) \rangle \rightarrow G \rightarrow G / \langle (a^2, e) \rangle \rightarrow e$$

Thus this sequence fits and is not split from above.

3. $C_2 \times C_2 \times C_2$

$$C_2 \cong \langle (a, e, e) \rangle$$

$$C_2 \times C_2 \times C_2 / \langle (a, e, e) \rangle = \{ \langle (a, e, e) \rangle, (e, b, e) \langle (a, e, e) \rangle, (e, e, c) \langle (a, e, e) \rangle, (e, b, c) \langle (a, e, e) \rangle \} \cong$$

$$C_2 \times C_2$$

$$e \rightarrow \langle (a, e, e) \rangle \rightarrow G \rightarrow G / \langle (a, e, e) \rangle \rightarrow e$$

Thus this sequence fits and is split by above.

4. $D_4 C_2 \cong \langle x^2 \rangle$

$$D_4 / \langle x^2 \rangle = \{ \langle x^2 \rangle, x \langle x^2 \rangle, y \langle x^2 \rangle, xy \langle x^2 \rangle \} \cong C_2 \times C_2$$

$$e \rightarrow \langle x^2 \rangle \rightarrow G \rightarrow G / \langle x^2 \rangle \rightarrow e$$

Thus this sequence fits and is not split from above.

5. $Q C_2 \cong \langle -1 \rangle$

$$Q / \langle -1 \rangle = \{ \langle -1 \rangle, i \langle -1 \rangle, j \langle -1 \rangle, k \langle -1 \rangle \} \cong C_2 \times C_2$$

Thus this sequence fits and is not split from above.

Problem 3

If G is split, there must exist a subgroup $D'_2 \cong D_2$ in G . This implies we can center all reflections and rotations at a single point, as there is a subgroup isomorphic to D_2 .

Let v be the smallest vector on the x -axis, and let w be the vector with the smallest height. Let z be the smallest vector in the lattice.

- If z lies on the x -axis, then $z = v$.
- If z lies on the y -axis, then $z = w$.

Assuming without loss of generality that z is in the first quadrant, we can act on it by r , so $r(z)$ is in our lattice. Then $r(z) + z$ lies on the x -axis. Since z is the smallest vector in the lattice, we have:

$$x(r(z) + z) < 2x(v) \Rightarrow 2x(z) < 2x(v)$$

If $2x(w) \neq x(v), 0$, then $z + r(z) - v$ would be a smaller vector than v on the x -axis, which is a contradiction.

Thus, if $x(z) = 0$, then $z = w$. If $x(z) = \frac{1}{2}x(v)$, then by similar reasoning, we find $x(z) = x(w)$. Since w has the smallest height and z is the smallest vector, it must be w . Therefore, our lattice is spanned by v and w .

We now have two cases: whether w is vertical or not.

1. **If w is vertical**, this corresponds to case 9:

$$G \cong \mathbb{Z} \times \mathbb{Z} \rtimes D_2 = \langle x, y, r, p_\pi \mid xy = yx, rp_\pi = p_\pi r, r^2 = e, p_\pi^2 = e, rxr = x, ryr = y^{-1}, p_\pi x p_\pi = x^{-1}, p_\pi y p_\pi = y^{-1} \rangle$$

with

$$\phi(r) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \phi(p_\pi) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

2. **If w is not vertical**, this corresponds to case 12:

$$G \cong \mathbb{Z} \times \mathbb{Z} \rtimes D_2 = \langle x, y, r, p_\pi \mid xy = yx, rp_\pi = p_\pi r, r^2 = e, p_\pi^2 = e, rxr = x, ryr = xy^{-1}, p_\pi x p_\pi = x^{-1}, p_\pi y p_\pi = y^{-1} \rangle$$

with

$$\phi(r) = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \quad \phi(p_\pi) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

If G is not split, then we have two cases:

1. If one reflection lifts to only a glide.

2. If both reflections lift to only a glide.

For the first case, if one reflection lifts to a glide, we make the glide parallel to the x -axis and the reflection parallel to the y -axis, centering the rotations at the origin. Let the min glide (meaning moves the minimum distance) parallel to the x -axis be defined as $t_{(a,h)}rt_{(0,-h)} = t_{(a,2h)}r$. Let a reflection parallel to the y -axis be $t_{(b,0)}r_y t_{(-b,0)} = t_{(2b,0)}r_y$.

Now, consider:

$$t_{(a,2h)}rt_{(a,2h)}r = t_{(a,2h)}t_{(a,-2h)}rr = t_{(2a,0)}$$

Thus, we have a vector on the x -axis. We call the minimal vector on the x -axis v . Assume $v = (c, 0)$ with $c < a$. Then consider:

$$t_{(-c,0)}t_{(a,2h)}r = t_{(a-c,2h)}r$$

Since $a - c < a$, this contradicts the minimal glide assumption, so $v = (2a, 0)$.

Similarly, we can find a vector on the y -axis by acting with r_y , and then finding the minimal vector, which we call w .

Let z be the smallest vector. If z lies on any axis, it must be v or w . If the x - or y -coordinate of z is zero, then it is either v or w , leaving us with $z = (a, 0.5y(w))$ using a similar argument as above.

Consider the element:

$$t_{-z}t_{(a,2h)}r = t_{(0,2h-0.5y(w))}r$$

Then:

$$t_{(0,2h-0.5y(w))}rt_{(0,2h-0.5y(w))}r = t_{(0,2h-0.5y(w))}t_{(0,-2h+0.5y(w))}rr = \text{id}$$

This is a reflection parallel to the horizontal axis, which is not allowed. Thus, this minimal vector z cannot exist. Therefore, the minimal vector is either v or w , so these two span the lattice.

Thus:

$$t_{(a,2h)}r = t_{\frac{1}{2}v+cw}r \in G \quad \text{for } c \in \mathbb{R}$$

and

$$t_{(2b,0)}r_y = t_{kv}r_y \in G \quad \text{for } k \in \mathbb{R}$$

Now, consider:

$$t_{kv}r_y t_{\frac{1}{2}v+cw}r = t_{(k-\frac{1}{2})v+cw}p\pi \in G$$

Thus:

$$t_{(k-\frac{1}{2})v+cw}p\pi p\pi = t_{(k-\frac{1}{2})v+cw} \in G$$

implying $k - \frac{1}{2} \in \mathbb{Z}$ and $c \in \mathbb{Z}$ since this translation is in my group, and we only have integer linear combinations of v and w .

So:

$$t_{-cw}t_{\frac{1}{2}v+cw}r = t_{\frac{1}{2}v}r = t_{(a,0)}r \in G$$

Also, since $k - \frac{1}{2} \in \mathbb{Z}$ and $2b = 2ak \Rightarrow b = ak \Rightarrow k = \frac{d}{2}$ for d odd, we have $b = \frac{ad}{2}$. Setting $d = 1$, we find $b = \frac{a}{2}$, so:

$$t_{(a,0)}r_y \in G$$

Thus, we have case 11:

$$G = \langle \mu, y, r, p\pi \mid \mu^2 y = y\mu^2, r^2 = p\pi^2 = e, r\mu r = \mu^{-1}, rp\pi = p\pi r, ry r = y, \mu y \mu^{-1}, r\mu^{-1} = \mu r, p\pi \mu^{-1} = \mu p\pi, p\pi \mu p\pi = \mu^{-1}, p\pi y p\pi = y^{-1} \rangle$$

If two reflections lift to glides, we pick the minimal glides $t_{(a,2c)}r$ and $t_{(2d,b)}r_y$ parallel to their respective axes. The minimal vectors on the x -axis and y -axis, as in the previous proof, are $v = (2a, 0)$ and $w = (0, 2b)$, respectively.

Then:

$$t_{(a,2c)}r = t_{\frac{1}{2}v+kw}r \quad \text{for } k \in \mathbb{R}$$

$$t_{(2d,b)}r_y = t_{jv+\frac{1}{2}w}r_y \quad \text{for } j \in \mathbb{R}$$

Now consider:

$$t_{jv+\frac{1}{2}w}r_y t_{\frac{1}{2}v+kw}r = t_{(\frac{1}{2}-j)v+(\frac{1}{2}+k)w}p\pi \in G$$

Thus:

$$t_{(\frac{1}{2}-j)v+(\frac{1}{2}+k)w}p\pi p\pi = t_{(\frac{1}{2}-j)v+(\frac{1}{2}+k)w} \in G$$

so we require $\frac{1}{2} - j \in \mathbb{Z}$ and $\frac{1}{2} + k \in \mathbb{Z}$.

Since $k + \frac{1}{2} \in \mathbb{Z}$ and $2c = 2ak$, it follows that $k = \frac{f}{2}$ for some odd integer f . Choosing $f = 1$, we get $c = \frac{a}{2}$. Similarly, $d = \frac{b}{2}$, so $t_{(a,0)}r \in G$ and $t_{(0,b)}r_y \in G$.

Thus, we have case 10:

$$G = \langle \mu, \gamma, p\pi \mid \mu^2 \gamma^2 = \gamma^2 \mu^2, \gamma \mu^{-1} = \mu \gamma^{-1}, \mu p\pi \mu^{-1} = \mu^2 p\pi, \mu \gamma^{-1} = \gamma \mu^{-1}, \gamma p\pi \gamma^{-1} = \gamma^2 p\pi \rangle$$

Problem 4

If G lifts one reflection to a reflection s.t. a rotation of order 4 is on that line; thus, make the reflection like the x -axis, let this reflection be r with the center of the rotation $p_{\pi/2}$ at the origin.

Let x be a vector. We can act on x by r to obtain something on the x -axis. Let v be the minimal vector on the x -axis.

Applying a rotation, we obtain a vector on the y -axis. Let w be the smallest vector on the y -axis.

Now, consider $p_{\pi/2}(v) - w$; this vector would be shorter than w , which is a contradiction. Thus, $p_{\pi/2}(v) = w$, meaning they must be the same vector.

Let z be the smallest vector and, without loss of generality, assume it lies in the first quadrant. By a similar argument as above, we either conclude that z is on one of the axes or that $z = (0.5 \cdot x(v), 0.5 \cdot x(v))$.

If this is our minimal vector, we can rotate by 45° to get a new coordinate system where the lattice is spanned by z and $p_{\pi/2}(z)$ or by v and w .

Thus, for any coord. system we will have a line of reflection with a center of an order 4 rotation.

Then, we can use $p_{\pi/2}$ on the reflection to get all reflections

Thus, this is a clear subgroup in G that is isomorphic to D_4 thus we are in the split case

Thus, with a square lattice we have case 13:

$$G \cong \mathbb{Z} \times \mathbb{Z} \rtimes D_4 = \langle x, y, r, p_{\pi/2} \mid xy = yx, rp_{\pi/2} = p_{\pi/2}^{-1}r, r^2 = e, p_{\pi/2}^4 = e, rxr = x, ryr = y^{-1}, p_{\pi/2}xp_{\pi/2}^{-1} = y, p_{\pi/2}yp_{\pi/2}^{-1} = x^{-1} \rangle$$

For:

$$\phi(r) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \phi(p_{\pi/2}) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

If G does not lift any reflection to a reflection s.t. a rotation of order 4 is on that line.

Thus, this is the non split case because we cannot get 4 reflections intersection at one point which is needed for a subgroup isomorphic to D_4 .

Thus, a reflection lifted to only glide, thus assume for contradiction that any glide is also not on a center of rotation of order 4.

Thus, 4 glides are on a

Problem 5